

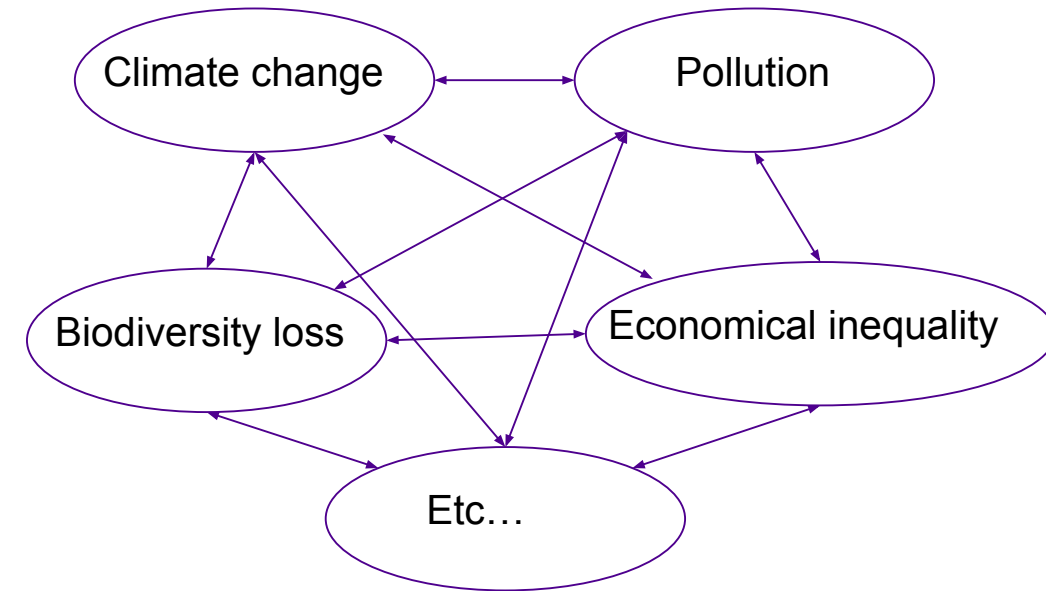
How do shared sustainability perceptions emerge in higher education settings?

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Motivation

- Complex, interconnected crises require sustainability competences and actions
- Higher-education institutions respond by developing sustainability education (Filho et al., 2024; Pietikäinen et al., 2024)
 - Multidisciplinary courses
 - Integrated sustainability goals
 - Degree programmes emphasizing sustainability



- **Problem:** sustainability is a broad and ambiguous concept
 - Perceptions on sustainability and sustainability action affect what kind of sustainability actors aim for and how (Salovaara et al., 2025; Veintie et al., 2026)
 - Diverse sustainability perceptions can advance or hinder sustainability education (Tziganuk & Gliedt, 2017; Aktas et al., 2015)

=> Evaluating the success of sustainability education requires understanding how & why sustainability perceptions change during studies

Research questions

How different actors' perceptions on sustainability and sustainability action change during higher education?

- 1) How the *concepts* used by the actors change during studies and when students start working?
- 2) How the *narratives* used by the actors change?
- 3) To what extent the actors share *common perceptions* and how this changes?
- 4) Does the *social network* of actors explain the changes?

Data collection

- Target degree programmes:
 - Technologies for Sustainable Use of Renewable Resources (University of Helsinki): Master's programme, 2 years
 - Natural Resource Technology (Jyväskylä University of Applied Sciences): Engineering programme, 4 years
 - Sustainability combined with technology and economy
 - ~50 students from various backgrounds, both Finnish and international
- Longitudinal data collection:
 - 3-5 interviews on perceptions on sustainability and sustainability actions
 - 8-16 questionnaires on social relationships
 - Pre-assignments written by students
 - Programme materials (curricula, course materials, ...)
- Data collection also from teaching staff and business collaborators



Sustainability science + narrative studies + network science

Why narratives?

- *Narrative* = causal-temporal structure that connects concepts, phenomena and actors, used to both make sense of the world and position oneself in it (Bruner, 1991; Björninen et al., 2020)
- Sustainability perceptions often include ideas of change or the need for one => perceptions often articulated through narratives
- *Master narratives* dominate the discourse (Hyvärinen, 2020)
 - Present in programme materials?
- *Counter-narratives* challenge the master ones
 - Created by students during studies?



Why networks?

- **Network** = a model of connections and interactions
 - Internet, social relationships, ecological systems, brain, ...
 - Network science offers mathematical and statistical tools for studying systems as networks
 - For details, see Newman, 2003
- Possible networks here:
 - Nodes = actors, edges = social relationships
 - Nodes = actors, edges = shared concepts and narratives => discourse networks (Leifeld, 2017)
 - Nodes = concepts, edges = co-occurrence => semantic networks (Shugars & Gonzáles-Bailón, 2022)

Methodology: Concepts & narratives

- *Concepts* identified through qualitative content analysis (inductive coding):
 - first coding round: different wordings of concepts recognized
 - second coding round: concepts recognized within and across subjects
- *Narratives* identified by close reading informed by narratology and sociolinguistics (Hatavara et al., 2020)
- Special attention on:
 - How complex are sustainability perceptions?
 - How and why they change?
 - Are there shared concepts and narratives across actors?

future-orientation

resource efficiency

life quality

For me Sustainability means using resources responsibly so that future generations can enjoy the same opportunities and quality of life that we have today. I have always believed that this world does not belong only to us. It also belongs to the next generations. Therefore, we have a responsibility to protect the environment and use resources wisely. My interest in sustainability developed during my bachelor's studies in Information and Communication Technology with a specialization in Bioeconomy, where I learned about renewable energy, circular economy principles, technologies and resource efficiency. My goal is to work in the renewable energy and sustainability sector, contributing to projects that promote clean energy, resource efficiency, waste reduction and circular economy practices. For me, sustainability is not just a career path but a commitment to creating a better future for present and future generations.

circular economy

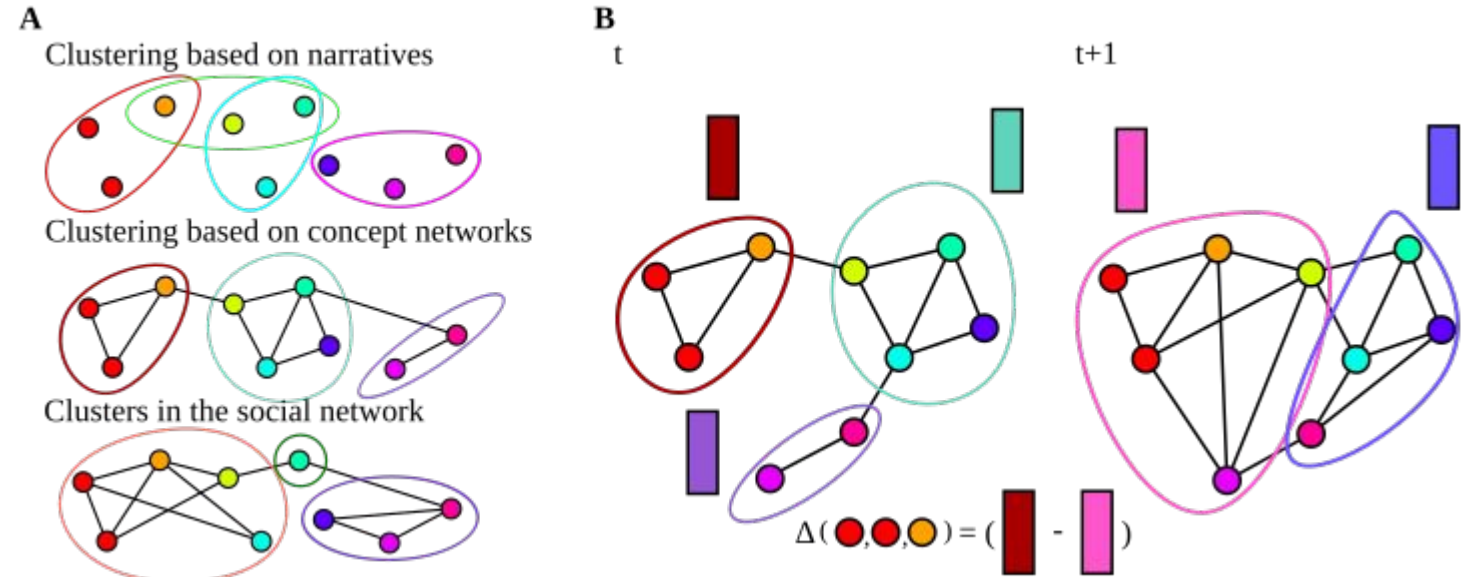
moral obligation

clean energy

Methodology: Networks

Shared sustainability perceptions

- Network construction:
 - Nodes = actors
 - Edges = shared concepts / social relationships
- Actor clustering based on:
 - Shared narratives
 - Network clustering methods
- Do the clusters overlap?
- Does background information explain clustering?
- How does the clustering change?
(We expect it to become less clear)



Sustainability perceptions in social networks

- Epidemic spreading & complex contagion
(Pastor-Satorras et al., 2015; Duanmu & Chai, 2025; Raponi et al., 2025)
 - Used to study spreading of diseases, innovations & rumours
- How do narratives spread in social network?

Preliminary results

- 12 pre-assignments written by the University of Helsinki students
- Brundtland definition of sustainability used in 4 responses + 4 other references to future or long-term
- Different categories of sustainability:
 - ecological + economic + social: 5
 - ecological: 3
 - ecological + economic: 2
 - ecological + social: 2
 - university staff in Veintie et al., 2026: ecological was the most common category but not included in all responses
- Non-human/"more than human" perspective in 2 responses only

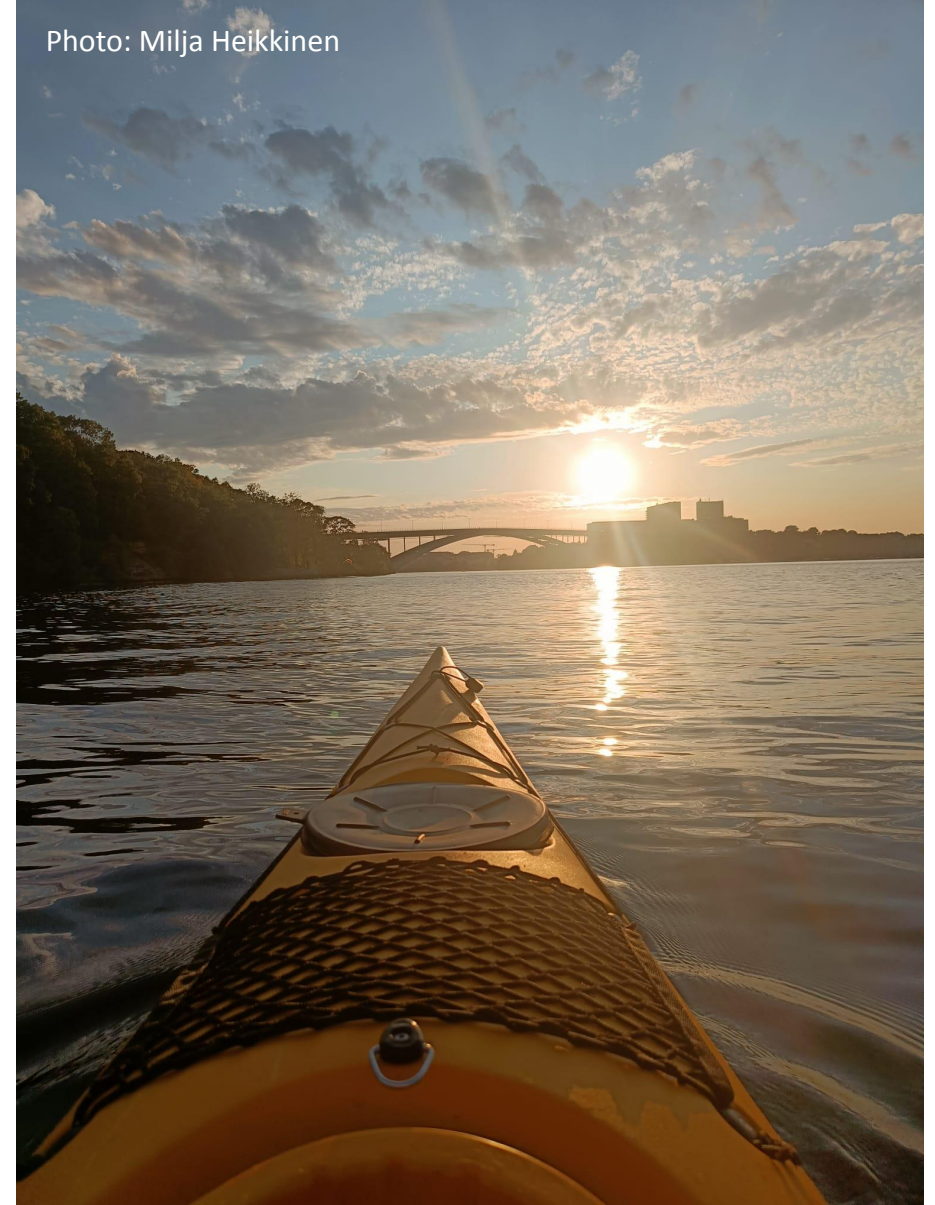
Photos: Milja Heikkinen
(featuring Riepu, Helmi & OK)



Expected impact

- New understanding on how sustainability perceptions change in higher education
 - Emergence of shared perceptions from the social and cultural setting of degree programmes?
 - Dynamics of master and counter-narratives?
- An interdisciplinary methodological framework: sustainability science + narrative studies + network science
- For the participants: possibility to understand their perceptions and reflect on their role as a sustainability actor

Photo: Milja Heikkinen



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Thank you!

Questions, comments?

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Slides:

